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Nicholas Walker

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EDUCATION

Rensselaer Polytechnic Institute (RPI) August 2021 – May 2025
Dual B.S. in Mechanical Engineering and Aerospace Engineering *Economics Minor*

- **Stanley Weiss Award** – 1st Place Capstone Design Award, May 2025
- Aerospace Specialty GPA: 3.6

EDUCATION

Computational Hypersonic Flow, RPI Troy, NY
Undergraduate Researcher *Jan 2025 – Ongoing*

- Investigating shock wave-boundary layer interactions (SWBLI) in hypersonic flow to assess effects on stability
- Developing computational models using OpenFOAM to analyze shock-induced separations with GMSH

Design Engineer, GE Vernova Schenectady, NY
Engineering Intern *Jan 2024 – April 2024*

- Led root-cause analyses of valve failures with CAD 3D models, identifying critical design flaws in PLM
- Analyzed last stage turbine part model and created detailed drawings of individual components with GD&T
- Facilitated comparative FFT analysis of in-house FEA software, revealing a 15% difference in results
- Prepared and delivered all technical reports and presentations, using various MS Office products

Structural Damage Diagnostic, RPI Troy, NY
Undergraduate Researcher *Sep 2023 – Dec 2023*

- Developed FEM/SEM-based structural health monitoring (SHM) models to assess Airbus UAV damage states
- Modeled structural behavior in FEA with Abaqus to analyze vibrational responses, fatigue, and failure modes
- Implemented damage state parameterization, processing sensor data in MATLAB achieving 95% result accuracy
- Synchronized LabVIEW system for signal generation and real-time data acquisition
- Calibrated sensors and DAQ data acquisition to convert signals to frequency domain for stress analysis

PROJECTS

Europa Orbiter, RPI *Sep 2024 – Dec 2024*

- Recognized with 1st Place Award for top overall capstone project at RPI across all engineering disciplines
- Designed and optimized the Minos Probe deployment mechanism from Europa Orbiter with a team of six
- Visualized system kinematics in Simulink, optimizing mass distribution, load paths, and thermal resilience
- Conducted orbital mechanics and trajectory analysis in AGI STK to ensure mission viability

Vertical Fin Array Design, RPI *May 2024 – August 2024*

- Designed heat sink using ICE9 Flex TPU to reduce heater temperature under natural convection
- Documented thermal evaluation using Rayleigh and Nusselt correlations to enhance fin efficiency by 15%

Yonka Rocket Design, RPI *Jan 2023 – May 2023*

- Led team in 1st place ranking RPI sponsored project to design and build carbon fiber rocket to reach Mach 2
- Modeled parts in Siemens NX, and performed dynamic and stress analysis using Femap with Nastran
- Coded Python script to optimize thrust and drag, increasing altitude performance by 30%

Automotive Wing Analysis, RPI *Jan 2023 – May 2023*

- Applied flow and boundary conditions in Altair CFD to support accurate aerodynamic post-processing
- Utilized HyperMesh to reduce simulation time by 40% while maintaining wing geometry accuracy

Fil-A-Bot Sustainability Device, RPI *Jan 2023 – May 2023*

- Designed and built a plastic to 3D printing filament recycling device with parts modeled in SolidWorks
- Programmed Arduino microcontrollers to regulate nozzle temperature and motor speed in real-time

SKILLS

Software Skills: Siemens NX, ANSYS, Nastran, SolidWorks, Autocad, Abaqus, 3Ds, Catia, Stk, LTSpice, Creo
Coding Languages: Matlab, Python, Linux, C++, Arduino